

MATH 126 SYLLABUS

1. GENERAL INFORMATION

Updated: Aug 22, 2026

- **Instructor:** Zhenhao Li (please feel free to call me Kevin).
- Best way to contact me is by email at zli1010@berkeley.edu.
- Lectures are on **Tuesdays** and **Thursdays** in **Dwinelle 215** from **2-3:30 pm** in **Evans 791**.
- Office hours will be Tuesday and Thursdays after class **3:30-5 pm**.
- All course materials will be made available on the class Canvas.

1.1. **Prerequisites.** Formal prerequisites for this course are a solid background in multivariable calculus, linear algebra, and ordinary differential equations at the level of **Math 53, 54**. However, this is an advanced mathematics course, and to properly follow the material, familiarity with real analysis and proof-writing at the level of **Math 104** is strongly recommended.

2. COURSE MATERIAL

The course provides a broad introduction to partial differential equations. Topics will include:

- (1) Introduction and review of basic concepts from calculus, ODEs, and analysis.
- (2) First order equations and the method of characteristics.
- (3) Solution and representation formulae for three basic PDEs: wave, heat, and Laplace's equation.
- (4) Separation of variables for wave and Laplace's equation.
- (5) Fourier series with applications to solving wave and heat equations.
- (6) Depending on interest and time, we may cover additional topics such as variational methods, elliptic regularity, nonlinear equations.

2.1. **References.** Lecture notes and readings will be posted after each class. The course will mainly follow

- *Introduction to Partial Differential Equations* by David Borthwick,

which you can download using SpringerLink access provided by Berkeley. For those interested, other good references include

- *Partial Differential Equations* by Lawrence C. Evans,
- *Partial Differential Equations* by Fritz John,
- *Partial Differential Equations, An Introduction* by Walter Strauss,
- *Partial Differential Equations: An Accessible Route through Theory and Applications* by András Vasy.

3. EXAMS AND GRADING

Your grade will be computed by using the best of the following schemes

- 15% homework, 25% midterm 1, 25% midterm 2, 35% final,
- 15% homework, 25% midterm 1, 60% final,
- 15% homework, 25% midterm 2, 60% final,
- 15% homework, 85% final.

3.1. **Exams.** You will have two in-class midterm exams:

- Midterm 1: Tuesday, **Oct 13**
- Midterm 2: Tuesday, **Nov 17**

You must take the final exam to pass the course, which will occur on

- Final exam: Tuesday, **Dec 15**, 8:00–11:00am.

3.2. **Homework.** There will be weekly problem sets, due **11:59pm on Thursdays**. The problem sets will either be graded on completion, or based on one problem drawn at random. They must be submitted electronically via Gradescope. You are highly encouraged to type the problem sets. It is your responsibility to make the homework legible.

No late assignments will be accepted, but two lowest homework grades will be dropped in the determination of grades.

You are required to give a collaboration and an AI disclosure on the problem set if you collaborated with anyone or consulted a generative AI for the problem set (e.g. ChatGPT, Gemini, Claude). Failure to do so will be treated as academic dishonesty. In other words, you must cite all the sources you use, list everyone you collaborated with, and give a description of how generative AI was used to help you on the problem set. However, I want to emphasize that collaboration and using generative AI to enhance your learning is highly encouraged.